many-valued

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June 24, 2024

1 All many-valued ideas

To denote the constructor of an uncertain formula, we will use the symbol Ð, because it is symmetrical (neither $\bot$, nor $\top$ are symmetrical). And it looks like crossed D, and "defined" starts with D, and it is connected with not being defined: normally(but not always!), if f is a functional symbol, $f(..., \text{undef}, ...)$.

$$I_{\text{Term}}(Ð) \coloneqq U$$

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Remark: actually we can use 6 truth values.
True = (present, stable(True))
False = (present, stable(False))
U1 = (absent, stable(True))
U2 = (absent, stable(False))
U3 = (present, unstable)
U4 = (absent, unstable)
Connectives are contagious in the first argument, it means the following:
Let's say # is an interpretation of a connective, and v is a truth value, then
(absent, ...) # v = (absent, ...)
(note, that "..." in lhs and rhs are not necessary the same)
(..., stable(True)) $\wedge$ (..., stable(True)) = (..., stable(True))
But two predicates can stabilize each other, so these truth values does not have a truth table, that's why we will not use them.

Also therefore Uncertain OR Uncertain is not necessary Uncertain in such semantics! Therefore we need to require even more: the certainity of every subformula, otherwise it will complicate the system too much?

e.g. $(f(x) = 5) \vee (x \notin \text{dom}(f))$.
Or will it require too much to require?
Yes, it will be too much, since then
(f function) $\wedge$ ($x \in \text{dom}(f)$) $\longrightarrow$ ($f(x) = f(x)$)
will not work normally.
Therefore we need something more interesting like:
(False $\longrightarrow$ Uncertain) := True.
Or even brave & better: reject the whole idea of having a defined truth table for uncertain values(or at least having a full/complete truth table). Such idea

was wrong, because it is not true even in completely classical logic of existence: the formula "$(x \in y)$" is not valid, so it has no any particular defined truth value.

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Idea: Truth values of predicates are subconglomerates of $(2\times)\mathcal{U} \times \cdots \times \mathcal{U}$ of values, where $I(P)$ is true.
"(Always)True" corresponds to the whole subconglomerate $(1\times)\mathcal{U} \times \cdots \times \mathcal{U}$.
"(Always)False" corresponds to the empty subconglomerate $(0\times)\emptyset \times \cdots \times \emptyset$, which is simply $\emptyset$.
"Uncertain" corresponds to any other subconglomerate.
All products here are given n times: n is a valency of the predicate symbol P.
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For example, the following predicates:

- (of valency 0):
  "$\bot$" corresponds to 0.("False")
  "$\top$" corresponds to 1.("True")
  "Ð" corresponds to 2.("Uncertain")
  (from now on, we will forget about "$2\times$" form the definition, for simplicity)
- (of valency 1):
  "(X unique)" corresponds to the collection of all conglomerate-semisingletones.
  "(X exists)" corresponds to the collection of all nonempty conglomerates.
  "(X (well-)defined)" corresponds to the collection of all nonempty conglomerate-semisingletones, i.e. to the collection of all conglomerate-singletones
  "empty" corresponds to the collection of empty conglomerates, i.e. just $\{\emptyset\}$.
- (of valency 2):
  "$(x \in y)$" corresponds to collection of all ordered pairs of nonempty conglomerates, where each class from first conglomerate is in each class of the second conglomerate.
  "$(x \notin y)$" corresponds to collection of all ordered pairs of nonempty conglomerates, where each class from first conglomerate is not in any class of the second conglomerate.

Definition 1.1. (Theorem) Theorem is a predicate which has the truth value "True".

Idea 2:

Actually, variables are ordered: $x_1, x_2, x_3, \ldots$. So we can consider the following correspondence evaluation of the formula $x_2 \in x_4$ via assignment function w, with

$$(1\times)\mathcal{U} \times \{w(x_2)\} \times \mathcal{U} \times \{w(x_4)\} \times \mathcal{U} \times \mathcal{U} \times \mathcal{U} \ldots,$$

where "infinite product" is abuse of notation, which means a collection of functions f from $\mathbb{N}_0$ to $\mathcal{P}\mathcal{U}$, such that $f(0) = 1$.